

Section A

Answer **all** the questions.

Write your answers in the spaces provided.

1. **Fig. 1.1** shows a ribbon diagram of a protein at the tertiary level of protein structure.

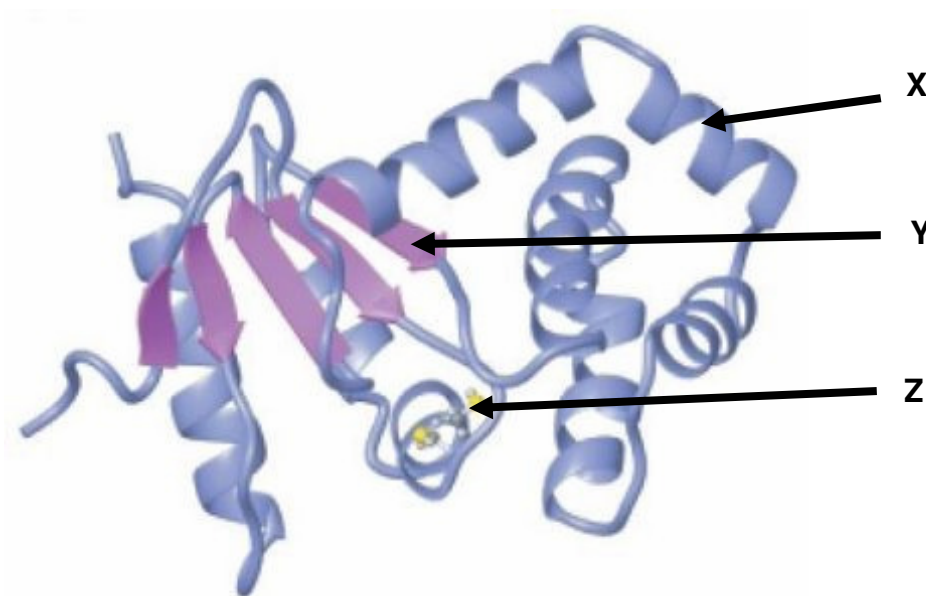


Fig. 1.1

- (a) Identify the type of secondary structure labelled **X** and **Y**.

X: _____

Y: _____ [1]

- (b) Identify a type of protein that would most likely have a 3-D configuration shown above.

_____ [1]

- (c) A group of tertiary proteins are twisted and folded such that there is a special functional site (labelled **Z** in **Fig 1.1**) within the structure. Name this site.

Site Z: _____ [1]

- (d) Site **Z** is involved in speeding up chemical reactions.

An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 seconds intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. The results are shown **Fig. 1.2**.

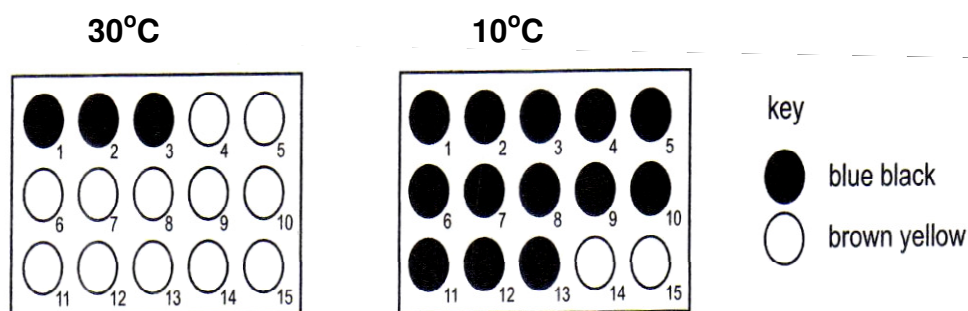


Fig. 1.2

- (i) Explain the results of the experiment as shown in **Fig. 1.2**.

[4]

- (ii) What role does Site **Z** have in the above reaction and how does changes in temperature affect site **Z**?

[3]

[Total:10]

2. Narcolepsy is a neurological condition characterized by excessive sleepiness in the day. It is an autosomal recessive condition. The condition has been documented in humans and several mammals, including dogs. The allele for normal level of wakefulness is denoted by **N** while that for narcolepsy is denoted by **n**. Another study concerned with pigmentation in dogs found that the allele coding for black (**B**) is dominant over that coding for brown (**b**).

A pure-breeding black dog with normal level of wakefulness is crossed with a true-breeding, brown, narcoleptic dog.

- (a) Write down the genotype of the F1 offspring from the cross stated.

_____ [2]

- (b) Two individuals of the F1 were subsequently mated to generate several litters of puppies. Assuming that the two genes are unlinked, perform a genetic cross in the space below to obtain the expected phenotypic ratio in the F2 generation. Show your workings clearly.

[4]

- (c) The total numbers of puppies and their phenotypes are recorded as below.

Black and non-narcoleptic	35
Black and narcoleptic	14
Brown and non-narcoleptic	10
Brown and narcoleptic	5

Given that the formula for chi-square below along with a section of the chi-square table, perform a chi-square test to show that the two genes are unlinked. (Show your workings clearly and in a detailed manner)

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O = the frequencies observed

E = the frequencies expected

$\sum$ = the 'sum of'

Degrees of freedom	Probability			
	0.95	0.50	0.10	0.05
1	0.004	0.46	2.71	3.84
2	0.10	1.39	4.60	5.99
3	0.35	2.37	6.25	7.82
4	0.71	3.36	7.78	9.49

4]

[Total:10]

3. The Lake of Tanganyika in Africa has been considered by evolutionary scientists as a type of evolutionary playground for a type of fish known as the cichlid, in a way similar to how the Galapagos Islands allowed for adaptive radiation in the finches described by Darwin. In this lake, over 250 species of cichlids have evolved through speciation within a relatively short span of over 10,000 years. Other than the very diverse coloration and morphology seen in this myriad of species found, individual species also differ widely in behaviour.

For some species (termed as mouthbrooders), dozens of hatched young are kept safe in the enlarged mouths of parents until the fries are relatively advanced. There are also egg robbers that force the parents to release some of the young to feed on them.

- (a) Explain how mouthbrooding behaviour in certain cichlid species might have evolved.

[4]

In a hypothetical scenario, evolutionary scientists isolated the DNA found in several species of cichlids, in an attempt to clarify their evolutionary relationship. A particular segment of DNA of a known gene was examined, in which there appears to be a high number of nucleotide differences between species.

	DNA sequence
Species A	A-T-C-T-G-G-A-T-C-A-G-G-G-C-C-C-G-G-A-T-T
Species B	A-T-G-A-G-G-T-T-C-A-G-G-A-C-C-G-G-G-A-T-T
Species C	A-T-C-A-G-G-T-T-C-A-C-G-A-C-C-G-G-G-A-T-T
Species D	A-T-C-T-G-G-A-T-C-A-G-G-G-C-C-C-G-G-A-T-A

- (b) In comparison to the DNA sequence in **species A**, note down the number of nucleotide differences that **species B, C, and D** have.

_____ [1]

- (c) Based on the DNA sequence presented, construct a cladogram showing the evolutionary relationship between the four species. (A cladogram is a branching, treelike, graphical representation of the phylogenetic relationships between organisms showing which species have branched from a common ancestor)

[2]

- (d) Explain three advantages in using molecular homology (especially DNA homology) in this scenario to clarify evolutionary relationships, in comparison to using anatomical or embryological homology.

_____ [3]

[Total: 10]

4. Restriction enzymes (also known as restriction endonucleases) are used to cut DNA molecules.

(a) State the source and natural use of restriction enzymes.

 [2]

The target sites for three restriction enzymes are shown in **Table 3.1**.

name of restriction enzyme	DNA base sequence of target site
A	$\begin{array}{c} \text{-C-C-C G-G-G-} \\ \text{-G-G-G C-C-C-} \end{array}$
B	$\begin{array}{c} \text{-C C-G-G-} \\ \text{-G-G-C C-} \end{array}$
C	$\begin{array}{c} \text{-A A-G-C-T-T-} \\ \text{-T-T-C-G-A A} \end{array}$

Table 3.1

Fig. 3.1 shows part of a plasmid, including details of the target site for restriction enzyme **A**.

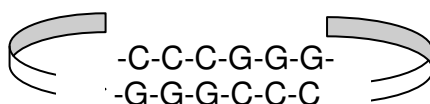


Fig. 3.1

(b) Explain whether it would be possible to cut this sequence using restriction enzymes **B** or **C**.

(i) enzyme **B**

 [1]

(ii) enzyme **C**

 [1]

Plasmids are loops of DNA, which are found in bacteria. They can be used to transfer human genes into bacterial cells. The human genes are inserted into the plasmids. In the transformation of recombinant plasmids into bacteria cells, three types of bacteria can be produced:

- A** – bacterial cell without plasmid
- B** – bacterial cell with plasmid but without gene of interest
- C** – bacterial cell with recombinant plasmid containing gene of interest.

(c) Suggest how bacterial **B** is possible.

[2]

(d) Using the plasmid illustrated in **Fig. 3.2**, explain how you would be able to select for bacterial **C**, assuming all three types of bacterial cells were present in the same culture medium.

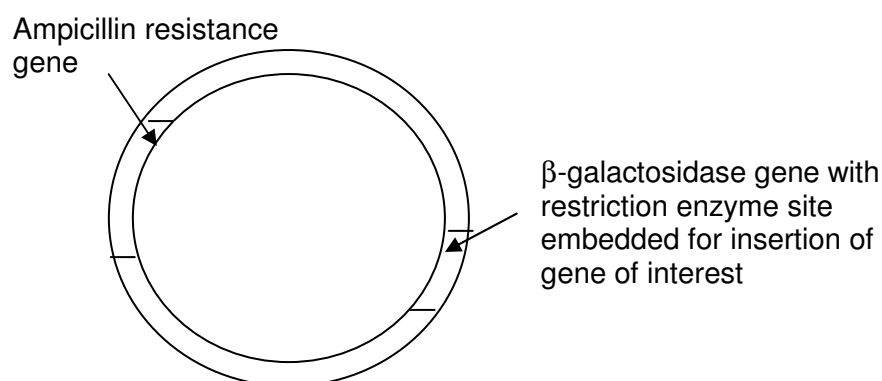


Fig. 3.2

[4]

[Total:10]

Section B

Answer only **one** question

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a)** Describe phosphorylation, giving examples wherever possible, indicate if this process occurs in photosynthesis, respiration or both. [4]
- (b)** Explain how the structure of mitochondria allows respiration to proceed. [6]
- (c)** Compare and contrast Krebs and Calvin cycles. [8]

[Total: 20]

- 6 (a)** Outline the differences between the prokaryotic control of gene expression with the eukaryotic model. [8]
- (b)** Outline the ways in which mutations could occur. [6]
- (c)** Describe how mutations might lead to cancer. [6]

[Total: 20]